

Uniform Integrability and the Vitali Convergence Theorem

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Introduction

When studying the various modes of convergence for sequences of functions, it is natural to harbor an expectation:

if f_n approaches f in ever stronger senses, shouldn't the integrals approach each other as well?

In other words, when can we pass from

$$f_n \rightarrow f$$

to

$$\int_X f_n d\mu \rightarrow \int_X f d\mu?$$

This question is a familiar one in Lebesgue integration theory. The monotone convergence theorem settles it by means of nonnegativity and monotonicity; the dominated convergence theorem instead requires a single integrable dominating function g with

$$|f_n| \leq g.$$

All of these conditions do the same job: they prevent the “mass” of the sequence from escaping in some way.

But the requirement of a dominating function is sometimes too strong.

A sequence of functions may admit no common integrable upper bound at all and still have perfectly good integral behavior. What really matters then is not any single dominating function, but a uniform control of the whole family over the “regions where the function values are large.”

This is **uniform integrability**.

1 Concentrated Mass

Let us start with the simplest example.

On $([0, 1], \mathcal{B}, m)$ define

$$f_n(x) = n\mathbf{1}_{(0, 1/n)}(x).$$

For any fixed $x > 0$, once n is large enough,

$$x \notin (0, 1/n),$$

and hence

$$f_n(x) \rightarrow 0.$$

Therefore

$$f_n \rightarrow 0 \quad \text{a.e.}$$

At the same time, for any $\varepsilon > 0$ and $n > \varepsilon$,

$$\{|f_n| > \varepsilon\} = (0, 1/n),$$

so that

$$m(\{|f_n| > \varepsilon\}) = \frac{1}{n} \rightarrow 0.$$

Hence the sequence also converges to 0 in measure.

However,

$$\int_0^1 |f_n| dx = n \cdot \frac{1}{n} = 1.$$

Thus

$$\|f_n\|_{L^1} = 1$$

stays constant, and a fortiori we cannot have

$$f_n \rightarrow 0 \quad \text{in } L^1.$$

What goes wrong?

It is not that the total integral of the sequence grows. In fact it is always equal to 1.

What actually changes is that this integral is squeezed onto smaller and smaller sets:

$$\text{height } n \quad \times \quad \text{width } \frac{1}{n}.$$

The sets shrink while the function values grow, and the two effects exactly cancel.

Consequently, merely knowing

$$\sup_n \|f_n\|_1 < \infty$$

is not enough.

We need to rule out this phenomenon of “a finite amount of mass concentrating in infinitely tall spikes.”

2 Definition

Let $(X, \mathcal{F}, \mu)$ be a measure space and let $\mathcal{G} \subset L^1(\mu)$ be a family of integrable functions.

If

$$\lim_{K \rightarrow \infty} \sup_{f \in \mathcal{G}} \int_{\{|f| > K\}} |f| \, d\mu = 0,$$

then $\mathcal{G}$ is said to be **uniformly integrable**.

For a sequence $\{f_n\}$ this reads

$$\lim_{K \rightarrow \infty} \sup_n \int_{\{|f_n| > K\}} |f_n| \, d\mu = 0.$$

The definition is concerned not with the whole integral

$$\int_X |f_n| \, d\mu,$$

but with the part where the function values are very large:

$$\int_{\{|f_n| > K\}} |f_n| \, d\mu.$$

For a fixed large K , each function may well have a part exceeding K . But if the family is uniformly integrable, then as soon as K is large enough, the integral mass carried by these parts can be made arbitrarily small **simultaneously**.

The crucial ingredient here is the

$$\sup_n$$

For any single $f_n \in L^1$ we already have

$$\int_{\{|f_n|>K\}} |f_n| d\mu \rightarrow 0.$$

Uniform integrability demands that this convergence hold uniformly over the whole family.

It is therefore not just another way of saying “each function is integrable,” but a genuine **property of the family**. In probability theory, uniform integrability is usually defined by exactly the same tail-integral condition.

3 Boundedness

First of all, uniform integrability implies

$$\sup_{f \in \mathcal{G}} \|f\|_1 < \infty.$$

The proof is not complicated.

By uniform integrability we can choose some $K > 0$ such that

$$\sup_{f \in \mathcal{G}} \int_{\{|f|>K\}} |f| d\mu < 1.$$

Split the integral into two parts:

$$\int_X |f| d\mu = \int_{\{|f| \leq K\}} |f| d\mu + \int_{\{|f| > K\}} |f| d\mu.$$

If $\mu(X) < \infty$, the first term satisfies

$$\int_{\{|f| \leq K\}} |f| d\mu \leq K\mu(X).$$

Hence

$$\|f\|_1 \leq K\mu(X) + 1.$$

Therefore

$$\sup_{f \in \mathcal{G}} \|f\|_1 < \infty.$$

The converse, however, fails.

The sequence

$$f_n = n\mathbf{1}_{(0,1/n)}$$

from the previous section is already a counterexample. Although

$$\sup_n \|f_n\|_1 = 1,$$

as soon as $n > K$ we have

$$\{|f_n| > K\} = (0, 1/n),$$

and so

$$\int_{\{|f_n| > K\}} |f_n| dx = 1.$$

Thus for every K ,

$$\sup_n \int_{\{|f_n| > K\}} |f_n| dx = 1.$$

The sequence is not uniformly integrable.

So

$\text{uniformly integrable} \implies L^1\text{-bounded}$

while the converse does not hold.

What separates the two is precisely control over the **concentration of mass**.

4 Small Sets

On a finite measure space, uniform integrability admits another very important interpretation.

If $\mathcal{G}$ is uniformly integrable, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every measurable set E with

$$\mu(E) < \delta$$

we have

$$\sup_{f \in \mathcal{G}} \int_E |f| d\mu < \varepsilon.$$

In words:

A sufficiently small set cannot carry a large amount of the L^1 mass of any function in the family.

The proof starts directly from truncation.

By uniform integrability, choose K such that

$$\sup_{f \in \mathcal{G}} \int_{\{|f| > K\}} |f| d\mu < \frac{\varepsilon}{2}.$$

For any measurable set E ,

$$\int_E |f| d\mu = \int_{E \cap \{|f| \leq K\}} |f| d\mu + \int_{E \cap \{|f| > K\}} |f| d\mu.$$

The first term is at most

$$K\mu(E),$$

and the second is at most

$$\frac{\varepsilon}{2}.$$

So it suffices to take

$$\delta = \frac{\varepsilon}{2K};$$

whenever $\mu(E) < \delta$ we then get

$$\int_E |f| d\mu < \varepsilon.$$

This is in fact closer to the geometric meaning of uniform integrability than the tail-integral definition.

Ordinary L^1 -boundedness only controls

$$\int_X |f|,$$

whereas uniform integrability further requires that **integral mass cannot concentrate arbitrarily on smaller and smaller sets.**

On a finite measure space, this “uniform absolute continuity on small sets,” together with uniform L^1 -boundedness, conversely characterizes uniform integrability.

5 L^p Control

Uniform integrability may still look somewhat abstract, but there is a very commonly used sufficient condition.

Let

$$1 < p < \infty$$

and suppose the family $\mathcal{G}$ satisfies

$$\sup_{f \in \mathcal{G}} \|f\|_p \leq C.$$

Then $\mathcal{G}$ is uniformly integrable (in the L^1 sense).

Indeed, on the set

$$A_K = \{|f| > K\}$$

we have

$$|f| \leq \frac{|f|^p}{K^{p-1}},$$

and hence

$$\int_{A_K} |f| d\mu \leq \frac{1}{K^{p-1}} \int_{A_K} |f|^p d\mu \leq \frac{\|f\|_p^p}{K^{p-1}}.$$

Therefore

$$\sup_{f \in \mathcal{G}} \int_{\{|f| > K\}} |f| d\mu \leq \frac{C^p}{K^{p-1}}.$$

Since $p > 1$,

$$K^{1-p} \rightarrow 0,$$

and uniform integrability follows.

This also shows why $p = 1$ is a special, critical position.

If all we have is

$$\sup_n \|f_n\|_1 < \infty,$$

the right-hand side produces no factor that vanishes as $K \rightarrow \infty$.

But as soon as there is even a little extra integrability,

$$p > 1,$$

we obtain the factor

$$K^{1-p},$$

which automatically controls the tail of large function values.

6 Dominated Convergence

The hypothesis of the dominated convergence theorem also yields uniform integrability automatically.

Suppose there exists

$$g \in L^1(\mu)$$

such that for all n ,

$$|f_n| \leq g \quad \text{a.e.}$$

If

$$|f_n| > K,$$

then necessarily also

$$g > K.$$

Hence

$$\{|f_n| > K\} \subseteq \{g > K\},$$

and

$$\int_{\{|f_n| > K\}} |f_n| d\mu \leq \int_{\{g > K\}} g d\mu.$$

Since $g \in L^1$,

$$\int_{\{g > K\}} g d\mu \rightarrow 0.$$

Therefore

$$\{f_n\}$$

is uniformly integrable.

Thus the common dominating function in the dominated convergence theorem actually provides more structure than is needed to prove convergence of the integrals.

What it gives is

$$|f_n| \leq g \in L^1,$$

whereas what is really responsible for controlling the tails of the integrals is

$\{f_n\}$ uniformly integrable.

This is one of the reasons why the Vitali convergence theorem is more flexible than the dominated convergence theorem. Uniform integrability can be viewed as a relaxation of the integrable-domination condition.

7 Vitali's Theorem

We can now return to the original question.

Suppose

$$\mu(X) < \infty$$

and

$$f_n \rightarrow f \quad \text{in measure.}$$

Convergence in measure alone does not imply L^1 convergence; the spike functions have already shown this.

The Vitali convergence theorem states that the missing condition is precisely uniform integrability:

$$f_n \rightarrow f \text{ in measure} + \{f_n\} \text{ uniformly integrable} \implies f_n \rightarrow f \text{ in } L^1.$$

In fact, within this framework one also obtains a corresponding converse: when $f_n \rightarrow f$ in measure, L^1 convergence and uniform integrability are essentially equivalent. The version of Vitali's theorem for probability measures is usually stated in exactly this form.

Why does the theorem hold?

Take $\eta > 0$ and define the bad set

$$A_n = \{|f_n - f| > \eta\}.$$

Since

$$f_n \rightarrow f \quad \text{in measure,}$$

we have

$$\mu(A_n) \rightarrow 0.$$

Now split the L^1 error as

$$\int_X |f_n - f| d\mu = \int_{X \setminus A_n} |f_n - f| d\mu + \int_{A_n} |f_n - f| d\mu.$$

On $X \setminus A_n$,

$$|f_n - f| \leq \eta,$$

so

$$\int_{X \setminus A_n} |f_n - f| d\mu \leq \eta \mu(X).$$

Meanwhile the measure of A_n tends to zero.

Uniform integrability guarantees exactly that on such shrinking sets,

$$\int_{A_n} |f_n|$$

as well as the corresponding contribution of the limit function cannot retain a fixed positive mass.

Hence the second term tends to zero as well.

Finally, letting first $n \rightarrow \infty$ and then $\eta \rightarrow 0$, we obtain

$$\|f_n - f\|_1 \rightarrow 0.$$

This proof reveals the true structure of Vitali's theorem:

convergence in measure		uniform integrability
$\Downarrow$	+	$\Downarrow$
measure of the bad set $\rightarrow 0$		small sets cannot carry much integral

The two pieces fit together into exactly L^1 convergence.

8 Two Kinds of Control

Looking back at the dominated convergence theorem from here, we find that the logic of the two theorems is in fact very close.

Dominated convergence uses one fixed function

$$g \in L^1$$

to control all the f_n at once.

This is **pointwise control**:

$$|f_n(x)| \leq g(x).$$

Uniform integrability does not require such a g to exist.

It only requires that the tails of the family be controlled uniformly in the integral sense:

$$\sup_n \int_{\{|f_n| > K\}} |f_n| \rightarrow 0.$$

This is **mass control**.

Consequently there can be uniformly integrable sequences for which no common L^1 dominating function exists. Vitali's theorem still handles such cases; this is exactly where it strictly goes beyond the direct scope of the dominated convergence theorem. Stanford's lecture notes on probability theory also give examples of this kind: "uniformly integrable, but with no common integrable dominating variable."

So the two theorems can be understood within one logical framework:

$$\text{DCT : } \text{common } L^1 \text{ domination} \implies \text{uniform integrability} \implies L^1 \text{ limit under control,}$$

and Vitali starts directly from the middle layer.

9 Infinite Measure

There is one more boundary here that is easy to overlook.

The finite-measure assumption above is not entirely inessential.

On

$$(\mathbb{R}, \mathcal{B}, m)$$

consider

$$f_n(x) = \frac{1}{n} \mathbf{1}_{[0,n]}(x).$$

Clearly

$$\|f_n\|_1 = 1.$$

Moreover, since

$$\|f_n\|_\infty = \frac{1}{n},$$

in fact

$$f_n \rightarrow 0$$

even uniformly, so of course also in measure.

On the other hand, for $K > 1$,

$$\{|f_n| > K\} = \emptyset$$

for all n , so by the tail-integral definition $\{f_n\}$ is uniformly integrable.

Yet

$$\|f_n\|_1 = 1$$

still does not tend to zero.

Here the mass does not concentrate on smaller and smaller sets; instead, a different phenomenon occurs:

the mass spreads out over larger and larger regions of the space.

On a finite measure space there is not enough room for this behavior; on an infinite measure space it requires additional control.

Hence, for a general measure space, controlling only the tail in the direction

$$|f_n| \rightarrow \infty$$

is not enough; one must also prevent the integral mass from escaping to the “far away” parts of the space.

General Vitali-type results therefore add a tightness condition: one can find a region of finite measure outside of which the integral mass is uniformly small for all n . The version for general measure spaces genuinely needs this spatial kind of control in addition to control of large values.

The two phenomena are best kept apart:

concentration :	mass squeezes into smaller and smaller sets,
escape :	mass runs off to larger and larger regions of space.

Uniform integrability rules out the former first; on infinite measure spaces the latter must be dealt with in addition.

Summary

Uniform integrability is not about whether a single function is integrable, but about whether the integral mass of a family of functions is uniformly controlled.

It requires

$$\lim_{K \rightarrow \infty} \sup_n \int_{\{|f_n| > K\}} |f_n| d\mu = 0,$$

that is, a sequence cannot preserve a fixed amount of integral mass through ever taller and ever narrower spikes.

Consequently,

$$\sup_n \|f_n\|_1 < \infty$$

by itself is not sufficient, whereas uniform L^p control for any $p > 1$ automatically yields uniform integrability.

More importantly, on a finite measure space the Vitali convergence theorem ties it to the convergence behavior of the sequence:

$$\boxed{f_n \rightarrow f \text{ in measure} \quad + \quad \{f_n\} \text{ uniformly integrable} \quad \implies \quad f_n \rightarrow f \text{ in } L^1.}$$

Convergence in measure tells us that the points where the error is large become fewer and fewer.

Uniform integrability tells us that these ever fewer points cannot still carry a large amount of integral.

The two conditions control two different problems, and together they supply exactly the missing step from convergence in measure to L^1 convergence.

From this point of view, uniform integrability is not yet another “mode of convergence.”

It is closer to a **compactness condition**: it controls how the mass of a family of functions is distributed, and decides when we may safely pass the limit inside the integral sign.